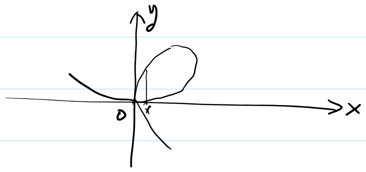
2025年3月10日 7:30

第7讲、隐函数微分法. 逆映射微分.

作业：P64. 2(1). 3(1). 4(1). 5.

例1.



$C^k(\Omega)$

$C^\infty(\Omega)$

设图中曲线是一个 C^1 类函数 $f(x,y)$ 的一条等高线. 则 $f(x,y)$ 在 $(0,0)$ 点梯度向量？

设 $\mathrm{grad} f(0,0) = \left(f_x'(0,0),\ f_y'(0,0) \right) \neq 0$

不妨设 $f_y'(0,0) \neq 0$, $f(0,0) = c$.

$f(x,y) = c$ $y = \varphi(x)$

$\Downarrow$

则 $\mathrm{grad} f(0,0) = \mathbf{0}$.

定理1. 设 $X = (x_1, x_2, \cdots, x_n)$. $Y = (y_1, y_2, \cdots, y_m)$

设 $n+m$ 元函数 $F_i(X, Y)$ $(i=1, 2, \cdots, m)$ 在 $(X_0, Y_0) \in \mathbb{R}^{n+m}$ 邻域 W 内有定义.

满足 ①. $F_i(X_0, Y_0) = 0$. $i=1, 2 \cdots m$

② $F_i \in C^k(W)$. $k \geq 1$.

③. $\left.\dfrac{\partial(F_1, F_2, \cdots, F_m)}{\partial(y_1, y_2, \cdots, y_m)}\right|_{(X_0, Y_0)}$ 可逆.

则. $\exists \delta > 0$, $\eta > 0$. s.t. $B(X_0, \delta) \times B(Y_0, \eta) \subset W$ 及

$G: B(X_0, \delta) \longrightarrow B(Y_0, \eta)$ 满足

1°. $Y_0 = G(X_0)$. 且 $\forall X \in B(X_0, \delta)$. $\exists ! Y \in B(Y_0, \eta)$. s.t. $F_i(X, Y) = 0$. $(i=1, 2 \cdots m)$

即 $Y = G(X)$ 是由 $F_i(X, Y) = 0$ 确定隐函数组.

2°. $G(X) \in C^k(B(X_0, \delta))$.

3°. $\forall X \in B(X_0, \delta)$.

$$\left.\frac{\partial(y_1, y_2, \cdots, y_m)}{\partial(x_1, x_2, \cdots, x_n)}\right|_X = -\left(\left.\frac{\partial(F_1, F_2 \cdots, F_m)}{\partial(y_1, y_2, \cdots y_m)}\right|_{(X,Y)}\right)^{-1} \cdot \left(\left.\frac{\partial(F_1, \cdots, F_m)}{\partial(x_1, \cdots, x_n)}\right|_{(X,Y)}\right)$$

$F(x,y) = 0$. $y'(x) = -\dfrac{F_x'}{F_y'}$

例2. 设 $\begin{cases} x = e^u \cos v \\ y = e^u \sin v \\ z = uv \end{cases}$. 求 $\dfrac{\partial z}{\partial x}$, $\dfrac{\partial z}{\partial y}$.

$$dz = \frac{\partial z}{\partial x} dx + \frac{\partial z}{\partial y} dy.$$

分析：$\begin{cases} F_1(x,y,z,u,v) = x - e^u\cos v \in C^1 \\ F_2(x,y,z,u,v) = y - e^u\sin v \in C^1 \\ F_3(x,y,z,u,v) = z - uv \in C^1 \end{cases}$

$$\det\frac{\partial(F_1,F_2,F_3)}{\partial(z,u,v)} = \det\begin{pmatrix} 0 & -e^u\cos v & e^u\sin v \\ 0 & -e^u\sin v & -e^u\cos v \\ 1 & -v & -u \end{pmatrix} = e^u \neq 0.$$

$\therefore \begin{cases} F_1=0 \\ F_2=0 \\ F_3=0 \end{cases}$ 确定隐函数组 $\begin{cases} z = z(x,y) \in C^1 \\ u = u(x,y) \in C^1 \\ v = v(x,y) \in C^1 \end{cases}$

二. 逆映射微分.

$y = f(x) \in C^1 \qquad y_0 = f(x_0).$

$F(x,y) = f(x) - y \qquad$ 若 $F_x'(x_0,y_0) \neq 0.$

$F(x,y) = 0. \implies x = \varphi(y) \in C^1 \qquad x_0 = \varphi(y_0)$

定理2. 设 $\Omega \subset \mathbb{R}^n$ 非空开集. 且映射 $Y = f(X): \Omega \to \mathbb{R}^n$ 连续可微.

若 f 在 X_0 的 Jacobi 阵 $Df(X_0)$ 可逆. $(X_0 \in \Omega)$. 记 $Y_0 = f(X_0)$.

则 $\exists\, \delta>0,\ \eta>0$. 及. 连续可微 $g: B(Y_0,\eta) \to B(X_0,\delta)$ 满足 $g(Y_0) = X_0$.

称 g 为 f 逆映射. 记 $g = f^{-1}$. 且 $\forall\, Y \in B(Y_0,\eta)$. $\quad \exists!\, X \in B(X_0,\delta)$.

$Df^{-1}(Y) = (Df(X))^{-1}. \qquad$ s.t. $Y = f(X)$.

证：$f: \Omega \to \mathbb{R}^n. \qquad f \in C^1(\Omega)$

$F(X,Y) = f(X) - Y \qquad F: \Omega \times \mathbb{R}^n \to \mathbb{R}^n$ 连续可微.

$Y_0 = f(X_0). \qquad \therefore F(X_0,Y_0) = 0.$

$$D_X F(X_0,Y_0) = \frac{\partial F}{\partial X}\bigg|_{(X_0,Y_0)} = Df(X_0) \text{ 可逆}.$$

$\therefore \exists\, \delta>0,\ \eta>0$. s.t. $\forall\, Y \in B(Y_0,\eta)$, $\exists!\, X \in B(X_0,\delta)$.

s.t. $F(X,Y) = 0$.

即 $F(X,Y)=0$ 确定隐函数组 $g: B(Y_0,\eta) \to B(X_0,\delta)$. 连续可微.

满足 $g(Y_0) = X_0. \qquad g(Y) = X, \qquad \forall\, Y \in B(Y_0,\eta).$

满足 $g(Y_0)=X_0$. $g(Y)=X$, $\forall Y\in B(Y_0,\eta)$.

$F(g(Y),Y)=0$. 即 $f(g(Y))=Y$.

$$\underset{f\circ g(Y)}{\overset{\|}{f(g(Y))}}\qquad f^{-1}(Y)=X$$

$\therefore g=f^{-1}$. 且 $f\circ f^{-1}(Y)=Y=I(Y)$

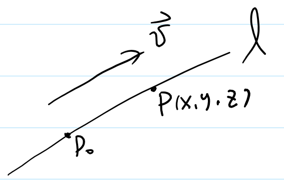
I 恒等变换: $\mathbb{R}^n\longrightarrow\mathbb{R}^n$.

$$Df(X)\cdot Df^{-1}(Y)=I_n \longrightarrow n\times n\text{阶单位阵}.$$

$$\therefore\ Df^{-1}(Y)=(Df(X))^{-1}.$$

三. 多元函数微分学应用.

l: $P_0(x_0,y_0,z_0)$, 方向向量 $\vec{v}=(v_1,v_2,v_3)$.



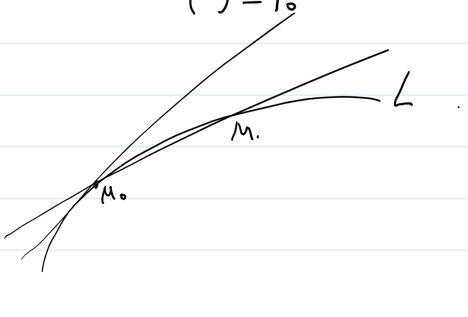
$\overrightarrow{P_0P}=t\vec{v}$. $t\in\mathbb{R}$.

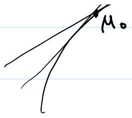
直线l参数方程: $\begin{cases} x=x_0+tv_1 \\ y=y_0+tv_2 \\ z=z_0+tv_3 \end{cases}$ $t\in\mathbb{R}$.

$$\frac{x-x_0}{v_1}=\frac{y-y_0}{v_2}=\frac{z-z_0}{v_3}\text{ —— 点向式方程}.$$

$v_1=0$. $\begin{cases} x=x_0. \\ \dfrac{y-y_0}{v_2}=\dfrac{z-z_0}{v_3} \end{cases}$

$v_1=0,\ v_2=0,$ $\begin{cases} x=x_0 \\ y=y_0 \end{cases}$





$L:\ \begin{cases} x = x(t) \\ y = y(t) \\ z = z(t) \end{cases}$ $t \in I$. 设 $x(t), y(t), z(t) \in C^1$.

$t_0 \in I$. $M_0(x_0, y_0, z_0)$ $\begin{cases} x_0 = x(t_0) \\ y_0 = y(t_0) \\ z_0 = z(t_0) \end{cases}$

$M \begin{cases} x(t_0+\Delta t) \\ y(t_0+\Delta t) \\ z(t_0+\Delta t) \end{cases}$ $\quad x(t_0+\Delta t) - x_0 = \Delta x$, $\quad y(t_0+\Delta t) - y_0 = \Delta y$, $\quad z(t_0+\Delta t) - z_0 = \Delta z$

$$\frac{x(t_0+\Delta t) - x(t_0)}{\Delta t} = \frac{\Delta x}{\Delta t} \ \downarrow\ x'(t_0)$$

$M(x_0+\Delta x,\ y_0+\Delta y,\ z_0+\Delta z)$

割线：M_0M $\quad \dfrac{x-x_0}{\Delta x} = \dfrac{y-y_0}{\Delta y} = \dfrac{z-z_0}{\Delta z}$

$$\frac{x-x_0}{\frac{\Delta x}{\Delta t}} = \frac{y-y_0}{\frac{\Delta y}{\Delta t}} = \frac{z-z_0}{\frac{\Delta z}{\Delta t}}$$

$\downarrow x'(t_0)$ $\quad \downarrow y'(t_0)$ $\quad \downarrow z'(t_0)$ $\quad (\Delta t \to 0)$

$\therefore$ L 在 M_0 切线：$\dfrac{x-x_0}{x'(t_0)} = \dfrac{y-y_0}{y'(t_0)} = \dfrac{z-z_0}{z'(t_0)}$

$\vec{\tau} = (x'(t_0),\ y'(t_0),\ z'(t_0))$ —— 为 L 在 M_0 切向量.

定义 1. 曲线上处处有非零切向量. 且切向量在曲线上连续变化.

称曲线是光滑的.

$L: \begin{cases} x = x(t) \\ y = y(t) \\ z = z(t) \end{cases}$ $\quad x(t), y(t), z(t) \in C^1$

且 $x'(t)^2 + y'(t)^2 + z'(t)^2 \neq 0$. 光滑.

1. 空间曲面切平面与法线.

设 $F(x,y,z)$ 在 M_0 邻域 W 内有定义. $F \in C^1(W)$.

$F(M_0) = 0$, $\quad F_z'(M_0) \neq 0$.

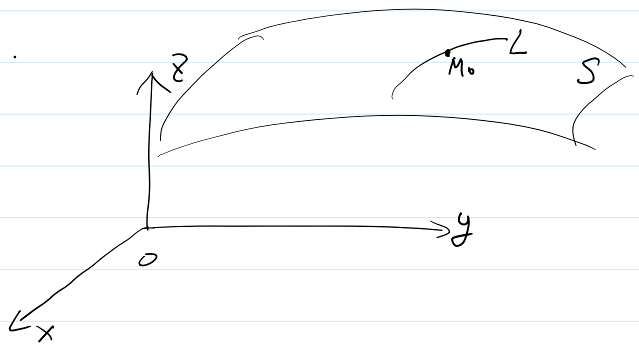
$F(x,y,z) = 0$. $\quad z = f(x,y)$

$F(x,y,z)=0.$ $\quad z=f(x,y)$

①. 曲面 S 一般方程 $F(x,y,z)=0.$

设 $\mathrm{grad}F(M_0)\neq 0.$ $\quad M_0(x_0,y_0,z_0).$ $\quad F$ 在 M_0 连续可微.

则 $F(x,y,z)=0$ 曲面.



任取 光滑曲线 L: $\begin{cases} x=x(t) \\ y=y(t) \\ z=z(t) \end{cases}$ $(t\in I)$ $\exists\, t_0\in I$ s.t. $\begin{cases} x_0=x(t_0) \\ y_0=y(t_0) \\ z_0=z(t_0). \end{cases}$

$$F(x(t),y(t),z(t))\equiv 0.$$

$$F_x'(M_0)\cdot x'(t_0)+F_y'(M_0)\,y'(t_0)+F_z'(M_0)\,z'(t_0)=0.$$

$$\mathrm{grad}F(M_0)\cdot\vec{\tau}=0$$

$$\vec{\tau}=(x'(t_0),\ y'(t_0),\ z'(t_0)).$$